

Monoclonal Immunoglobulin–Related Glomerular Diseases and Renal Amyloidosis

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MONOCLONAL GAMMOPATHY OF RENAL/CLINICAL SIGNIFICANCE

The glomerulus is a favorite site for the deposition of aggregated, misfolded, or abnormal proteins. Among these, monoclonal immunoglobulins (Igs) are increasingly recognized as a major cause of glomerular disorders (see also [Chapter 22](#)). Monoclonal Ig is produced by an abnormal B-cell clone, which most commonly does not cause cancerous symptoms but is quiescent or indolent, corresponding to a monoclonal gammopathy of undetermined significance (MGUS) or to an indolent lymphocytic or plasmacytic disorder. The association of such a “dangerous small B-cell clone” with related kidney manifestations is now recognized as monoclonal gammopathy of renal significance (MGRS).¹ Recently, the concept has been extended to monoclonal gammopathy of clinical significance (MGCS) to encompass the involvement of other organs.² Importantly, in patients with MGR(C)S-related organ lesions, current treatment is based on targeting the underlying B-cell clone, although it is not malignant per se.

Overt hematologic malignancies, such as symptomatic myeloma, B-cell lymphoma, or Waldenström macroglobulinemia (WM), may be complicated by the same kidney diseases as in MGR(C)S. More frequently, kidney disorders occurring during these malignant lymphoid and/or plasmacytic proliferations are related to tumor mass, including light chain (LC) cast nephropathy, kidney infiltration by malignant cells, and intracapillary thrombi of monoclonal IgM.^{3,4} In these cases, treatment is directed at the hematologic malignancy.

Kidney disorders related to MGRS are classified into three categories according to the composition and appearance of deposits by immunofluorescence (IF) and electron microscopy (EM), respectively ([Table 28.1](#)).⁵ The first category includes diseases with deposition of monoclonal Ig organized ultrastructurally into fibrils (immunoglobulinic amyloidosis) or microtubules (cryoglobulinemic glomerulonephritis [GN] and immunotactoid GN). The second category includes disorders characterized by nonorganized electron-dense granular monoclonal Ig deposits (i.e., monoclonal Ig deposition disease) and proliferative GN with monoclonal Ig deposits. In the third category, encompassing C3 glomerulopathy and thrombotic microangiopathy, no Ig deposits are present and glomerular lesions are thought to derive from local or systemic complement activation by the monoclonal Ig. This chapter focuses on glomerular diseases with organized and nonorganized monoclonal Ig deposits, except for cryoglobulinemic GN ([Chapter 22](#)), fibrillary GN ([Chapter 29](#)), monoclonal Ig-associated C3 glomerulopathy ([Chapter 23](#)), and renal thrombotic microangiopathy ([Chapter 30](#)).

Pathophysiology

The large spectrum of MGR(C)S-related diseases reflects the diversity of pathogenic mechanisms and affected tissues or organs. Kidney lesions most commonly result from deposition of all or part of the monoclonal immunoglobulins as aggregates, amorphous, crystalline, microtubular, or fibrillar forms. Other identified mechanisms include autoantibody activity directed against a tissue antigen or resulting in the deposition of immune complexes and complement activation.^{2,5,6} Of note, monoclonal Ig-mediated activation of the alternative pathway of the complement system has been documented recently in monoclonal Ig-associated C3 glomerulopathy (see [Chapter 23](#)).⁷

Only a minority of monoclonal immunoglobulins produced in clonal B-cell disorders are nephrotoxic. Pathogenicity appears mostly due to sequence peculiarities in the variable (V) domain that govern monoclonal Ig toxicity, including their propensity to aggregate and deposit.⁶ In light chain amyloidosis (AL), there is a strong prevalence for λ LC isotype, and two germline V λ genes are implicated in half of λ -LC-associated AL amyloidoses.⁸ In light chain deposition disease (LCDD), κ LC isotypes are predominant, with overrepresentation of the V κ 4-1 gene. The LC V domain features frequent glycosylation, hydrophobic residues, and high isoelectric point (pI) that could account for propensity to deposit along the anionic proteoglycans of the basement membranes.^{6,9}

Diagnostic Considerations

In any patient with proteinuria, serum and urine protein electrophoresis must be performed. In addition, serum and urine immunofixation and serum free light chain (FLC) assay should be considered, particularly in the 40- to 80-year age range.^{1,2,5}

In Patients With Known Monoclonal Gammopathy and Kidney Manifestations

The first step is to rule out symptomatic multiple myeloma (MM), WM, chronic lymphocytic leukemia (CLL), or other B-cell lymphoma, which would per se require specific hematologic treatment.

In patients with monoclonal gammopathy and no evidence of symptomatic hematologic disease, the diagnosis of MGRS-related kidney disease should be considered.

Regarding kidney disease, immunologic investigations should include serum complement studies and tests for cryoglobulins. In the absence of contraindications, a kidney biopsy should be performed in most patients. Exceptions are AL amyloidosis already proven on an extrarenal biopsy (see later), and any condition including age and comorbidities that would question the benefits of chemotherapy. If a concurrent cause of kidney disease such as diabetes is present, kidney biopsy should be considered only if presentation or clinical course is

TABLE 28.1 Classification of Glomerular Disorders With Monoclonal Immunoglobulin Deposits and Comparison With Fibrillary Glomerulonephritis

Glomerular Disease	Renal Symptoms	Light Microscopic Findings	Immunohistology	Ultrastructural Findings	Extrarenal Involvement	Identification of a Monoclonal Immunoglobulin	Hematologic and Immunologic Characteristics
Organized Monoclonal Ig Deposits							
Amyloid: AL, AH, AHL	Proteinuria, NS CKD Hypertension and hematuria uncommon	Congo red–positive mesangial and CW deposits Vascular and tubulointerstitial deposits common	AL: LC deposits (mostly λ) AH: HC deposits ($\gamma 1$, or $\gamma 4$, or α), with CH1 deletion AHL: LC and HC deposits	7–14 nm fibrils randomly arranged	Frequent: heart, liver, peripheral nerves	EP/IFX: 70%–90% FLC: 80%–90%	MGR(C)S Symptomatic MM uncommon WM
Immunotactoid glomerulopathy/ glomerulonephritis with organized microtubular immunoglobulin deposits	Proteinuria, NS CKD Microhematuria Hypertension	Mesangial proliferative GN Atypical MN MPGN pattern Interstitial tumoral infiltrate common (CLL)	Granular/smudgy deposits in mesangium and CW (predominantly subepithelial). Monotypic IgG deposits (IgG ₁ most common) ($\kappa > \lambda$) C3, C4, C1q deposits	10–60 nm microtubules with hollow core, arranged in parallel; similar microtubules in leukemic lymphocytes (if present)	Rare (peripheral nerve, skin)	EP/IFX: 40%–70% FLC: 20%	CLL (common) B-cell lymphoma MGRS MM Hypocomp. ~30%
Type I cryoglobulinemic GN	Proteinuria, NS CKD Microhematuria Hypertension Possible nephritic syndrome, AKI	MPGN pattern Endocapillary GN Glomerular thrombi common Intrarenal vasculitis occasional	Deposits in mesangium, CW (predominantly subendothelial), vascular walls. Glomerular thrombi IgG, IgM, or IgA ($\kappa > \lambda$) C3, C4, C1q deposits	10–90 nm microtubules with hollow core Extra + intracellular crystals (crystal-cryoglobulinemia)	Frequent: skin ++, peripheral nerve, joints	EP/IFX: 80% FLC: unknown	MGR(C)S MM B-cell lymphoma WM Hypocomp. common
Nonorganized Monoclonal Ig Deposits							
Monoclonal Ig deposition disease	Proteinuria, NS CKD Microhematuria Hypertension	Nodular glomerulosclerosis (constant in HCDD) Less often mesangial proliferative GN, crescents rare (<10%) and focal Thickened TBM and vascular walls	Linear deposits along TBM, GBM, and arteriolar myocytes LCDD: LC deposits only (mostly κ) HCDD: HC deposits only (γ or α), with CH1 deletion LHCDD: LC + HC deposits	Amorphous, finely granular deposits in TBM, GBM, mesangium, and arteriolar walls	Frequent (often oligosymptomatic): heart, liver, peripheral nerve	EP/IFX: 70%–100% FLC: ~100%	MGR(C)S Symptomatic MM WM Hypocomp. common in $\gamma 1$ and $\gamma 3$ HCDD

Glomerular Disease	Renal Symptoms	Light Microscopic Findings	Immunohistology	Ultrastructural Findings	Extrarenal Involvement	Identification of a Monoclonal Immunoglobulin	Hematologic and Immunologic Characteristics
Proliferative GN with monoclonal Ig deposits	Proteinuria, NS CKD Microhematuria Hypertension	MPGN pattern Endocapillary GN Membranous GN Mesangial proliferative GN	Granular mesangial and CW deposits IgG deposits (mostly IgG3), ($\kappa > \lambda$), rarely IgM, IgA, or LC alone C3 + C1q deposits	Nonorganized granular deposits in mesangium, subendothelial, and/or subepithelial zone	None	EP/IFX: 30% FLC: unknown	Usually none MGRS MM, B-cell lymphoma WM rare Hypocomp. ~30%
Fibrillary GN (See Chapter 29): Not Related to Monoclonal Ig Deposits							
Fibrillary GN	Proteinuria, NS CKD Microhematuria Hypertension Acute nephritic syndrome rare	Mesangial proliferative GN MPGN pattern Crescents in 10–20% Positive DNAJB9 staining in 100% Congo red negative ^a	IgG deposits (almost always polyclonal, mostly IgG4) ^b C3 ± C4, C1q	12–24 nm fibrils randomly arranged	None	EP/IFX <10% (fortuitous association in most cases)	HCV, autoimmune diseases, nonhematologic malignancy Hematologic disease very rare

^aRare fibrillary GN cases with Congo red weakly positive glomerular deposits have been described.⁴⁶

^bRare cases of IgG-negative fibrillary GN have been described.⁴⁷

AL, Amyloid with immunoglobulin heavy chains; AL, amyloid with immunoglobulin light chains; AKI, acute kidney injury; CLL, chronic lymphocytic leukemia; CKD, chronic kidney disease; CW, glomerular capillary walls; DNAJB9, DnaJ homolog subfamily B member 9; EP/IFX, electrophoresis/immunofixation; FLC, serum free light chains; GN, glomerulonephritis; HC, immunoglobulin heavy chains; HCDD, heavy chain deposition disease; Hypocomp., hypocomplementemia; Ig, immunoglobulin; LC, immunoglobulin light chains; LCDD, light chain deposition disease; LHCDD, light and heavy chain deposition disease; ITGP, immunotactoid glomerulopathy; MGR(C)S, monoclonal gammopathy of renal (clinical) significance; MM, multiple myeloma; MN, membranous nephropathy; MPGN, membranoproliferative glomerulonephritis; NS, nephrotic syndrome; TBM, tubular basement membrane; WM, Waldenström macroglobulinemia.

atypical. Importantly, when a kidney biopsy is performed, a sample should be processed for EM. Light microscopy (LM) should include Congo red staining. Immunofluorescence studies are crucial to determine the composition of deposits, to assess their monotypic nature and their concordance with the circulating monoclonal Ig. They should be performed using antibodies specific for κ and λ LCs, and, when deposits are composed of an entire IgG, heavy chain subclass typing is recommended. Except for most cases of AL amyloidosis, EM is required to assess the diagnosis of MGRS-related lesions to demonstrate the distribution and ultrastructural pattern of organization of deposits. When the diagnosis remains uncertain, additional techniques including paraffin IF after pronase digestion, proteomic analysis, and immuno-EM are required, preferentially after expert advice from a reference center.⁵

Careful attention should be paid to extrarenal manifestations. Tests for liver function and cardiac biomarkers (NT-proBNP and troponin) are mandatory.

Hematologic workup is the cornerstone of the management of MGR(C)S. It must include bone marrow examination with flow cytometric characterization of lymphocytic and plasmacytic populations and serum FLC (sFLC) measurement. Interpretation of FLC assays may be difficult in patients with impaired kidney function. Physiologically, FLCs undergo renal elimination, and the level of polyclonal κ and λ FLCs increases as glomerular filtration rate decreases. With the Freelite test, the “normal” FLC ratio is 0.26 to 1.65 in patients with preserved kidney function but is 0.34 to 3.10 in patients with severe kidney failure (chronic kidney disease [CKD] stage 5 or greater). Of note, several different assays for measuring FLC are now available but results from different assays cannot be directly compared so the same FLC assay must be used to monitor a particular patient throughout follow-up. The nature of the clone is usually in accordance with the isotype of the monoclonal Ig: plasmacytic when the monoclonal Ig is IgG, IgA, or LC only; and lymphoplasmacytic or lymphocytic when it is IgM.^{1,2,5,10} Exceptions exist, such as plasma cell clones harboring translocation t(11;14) and producing monoclonal IgM. Considering the high prevalence of monoclonal immunoglobulins after age 50 years (>3%), it is crucial to exclude a chance association between the kidney disease and the monoclonal gammopathy and obtain support for a causal link between the two. For example, demonstrating tissue deposition or precipitation of all or part of the circulating monoclonal Ig is a strong argument for clone toxicity. In the absence of such evidence, an indication for chemotherapy is more uncertain and should be validated by multidisciplinary evaluation.^{1,2,5,10}

In Patients Without Known Monoclonal Gammopathy but Monotypic Deposits/Inclusions or C3 Deposits/Thrombotic Microangiopathy on Kidney Biopsy

In this situation, the main concern is detecting and characterizing the pathogenic clone. The monotypic nature of renal deposits should be carefully assessed. Detailed immunologic hematologic investigations are required to identify a small monoclonal Ig, including serum and urine immunofixation, measurement of sFLC, and sensitive techniques (flow cytometry) to detect a small blood or bone marrow B-cell population.^{5,10}

Principles of Treatment

General Considerations

In patients with MGRS-related glomerulopathies, targeting the underlying B-cell clone is the only therapeutic option and requires collaboration between nephrologists and hematologists. Treatment is based on chemotherapy, either conventional or with high-dose melphalan (HDM) supported by autologous stem cell transplantation (ASCT), and anti-B-cell monoclonal antibodies (Mabs).^{2,10}

No strategy is available to limit monoclonal Ig toxicity by reducing its secretion, inhibiting its deposition, or other interventions. In AL amyloidosis, approaches aimed at eliminating amyloid deposits have been developed, using antibodies directed against conformational fibril epitopes, or agents targeting amyloid P component. However, none has been approved yet for use.^{8,11,12}

The choice of chemotherapy should consider kidney metabolism of drugs and their potential renal and extrarenal toxicity. Bortezomib can be used without dose adjustment, even in patients with end-stage kidney disease (ESKD). Thus, bortezomib-based regimens are often recommended because of their favorable benefit/risk profile.¹³ HDM/ASCT is feasible with acceptable toxicity in carefully selected patients with kidney failure due to AL amyloid or monoclonal Ig deposition disease (MIDD).^{8-10,14} The administration of rituximab raises no concerns in patients with reduced glomerular filtration rate (GFR). Current experience regarding the use of antiplasma cell Mabs, such as anti-CD38 Mabs in AL amyloidosis, is encouraging.^{8,15}

Indication for treatment and its modalities rely upon a benefit-to-risk approach considering natural history of kidney disease, extrarenal involvement, and comorbidities. It is reasonable to consider polychemotherapy and even HDM/ASCT in a patient with MGR(C)S and potential life-threatening clone-related lesions. In contrast, these strategies are not appropriate in a frail patient with MGRS and kidney-limited disease who already progressed to ESKD and is not eligible for kidney transplantation. In patients on maintenance dialysis, chemotherapy should be considered if extrarenal manifestations are present or if the patient is a candidate for kidney transplantation, given the high recurrence rate of MGRS-related kidney diseases in the allograft (see later).¹⁰

Treatment should be adapted to the nature of the clone. If the clone is plasmacytic, treatment is based on antimyeloma agents, particularly bortezomib-based regimens. Anti-CD38 Mabs are increasingly used in all MGR(C)S types. Treatment with an anti-CD20 Mab, such as rituximab, is usually not appropriate in patients with a plasmacytic clone, whose cells are commonly CD20 negative. If the clone is lymphoplasmacytic or corresponds to CLL or B-cell lymphoma, treatment should be adapted accordingly, usually based on anti-CD20 Mabs. The place of new agents, such as Bruton kinase inhibitors, remains to be assessed.^{2,5,8,10}

Evaluation of Response to Treatment

Achievement of the best hematologic response is the goal of treatment and the primary criterion for determining its optimal duration.^{2,5,8,10,16} In the most frequent types of MGR(C)S, AL amyloidosis, and MIDD, the quality of hematologic response is the main prognostic factor for both kidney and patient survival.^{8,9,13-18}

Accordingly, assessment of hematologic response to treatment is crucial. It depends on repeat evaluations of the pathogenic monoclonal Ig component, usually based on FLC variations. International response criteria have been validated in AL amyloidosis (see later).^{8,16} The validity of these criteria in MIDD is currently under evaluation. When the causal monoclonal Ig is not detectable or difficult to measure, evaluating the hematologic response is challenging. Repeated bone marrow examination, including molecular studies, may be considered.¹⁰

For kidney (and extrarenal) outcomes, hematologic response is necessary but not sufficient. The probability of improving organ function is also governed by the severity and chronicity of tissue damage. When a complete or near complete hematologic response has been achieved, chemotherapy should be stopped even if kidney symptoms persist, because organ response is usually delayed. For example, in AL amyloidosis, remission of nephrotic syndrome may require up to 1 year after hematologic remission.¹⁴ At least

partial regression of monoclonal Ig deposits and associated lesions after deep hematologic response has been documented histologically in animal models and in few patients with MIDD or AL amyloidosis.^{6,19}

Symptomatic Measures and Kidney Replacement Therapies

Supportive measures in MGRS-related glomerulopathies are generally similar to that recommended in glomerular diseases (see [Chapter 82](#)). In AL amyloidosis, blockers of the renin-angiotensin system and anticoagulants should be used cautiously due to the risk of severe hypotension and bleeding, respectively. Monitoring of side effects of chemotherapy includes prophylactic antibiotics.^{8,10}

Indications and modalities of chronic dialysis are similar to that in patients with other causes of ESKD. Patients with AL amyloidosis are more frequently exposed to complications due to multisystemic organ involvement.^{8,10,20}

Kidney transplantation may be proposed to selected patients with MGC(R)S without severe extrarenal disease. Transplantation should be performed preferentially in those who have achieved sustained deep hematologic remission (for at least 6 months) prior to the procedure.^{9,21–23} Recurrence occurs after variable delay depending on the type of kidney disease, leading to rapid graft loss in proliferative GN with monoclonal Ig deposits (PGNMID),²³ or after several years in AL amyloidosis.^{21,22} The recurrence rate is mostly determined by the quality of hematologic response prior to transplantation. In AL amyloidosis and MIDD, achievement of very good partial hematologic response or more is associated with acceptable long-term allograft and patient outcomes.^{9,21,22} Compared with the broader population of kidney transplant recipients, recipients with AL amyloidosis or MIDD are exposed to an increased risk of neoplastic and infectious complications.^{9,21} In contrast, as in all patients with MGRS, the risk of progression to overt symptomatic myeloma is low.^{9,21–23}

GLOMERULOPATHIES WITH ORGANIZED MONOCLONAL IMMUNOGLOBULIN DEPOSITS

On EM examination, glomerular monoclonal Ig deposits may be organized into fibrils, crystals, or microtubules.

Fibrillar Deposits

Fibrillar deposits nearly always correspond to LC (AL) amyloidosis, which is the main type of renal AL amyloidosis (see later). Nonamyloid fibrillary GN is nearly always characterized by deposition of polyclonal IgG (see [Chapter 29](#)).^{5,24,25}

Crystalline Deposits

Type I cryoglobulins (usually IgG) sometimes display a crystalline pattern defining crystalglobulinemia.⁵ LC crystalline inclusions within glomerular cells have been rarely described, mostly in patients with Fanconi syndrome or crystal-storing histiocytosis⁵ (see [Chapter 50](#)).

Microtubular Deposits

Microtubular deposits are observed in type I and type II cryoglobulinemic GN (see [Chapter 22](#)).⁵ They also characterize immunotactoid glomerulopathy, also referred to as GN with organized microtubular monoclonal Ig deposits (GOMMID).^{24–27}

IMMUNOTACTOID GLOMERULOPATHY (GOMMID)

Immunotactoid glomerulopathy (ITGP/GOMMID) is extremely rare, representing less than 0.05% of all kidney biopsies.^{25,26} Diagnosis is

usually made between age 50 and 60 years. Kidney manifestations include proteinuria often of nephrotic range, microscopic hematuria, and mild to severe CKD.^{24–27} Extrarenal deposits in the skin or peripheral nerves have been very rarely reported.²⁴

Histologically, ITGP/GOMMID usually shows atypical membranous glomerulopathy with segmental mesangial proliferation, or a membranoproliferative GN (MPGN) pattern. Interstitial infiltration with clonal lymphocytes is common in patients with an underlying CLL. On IF, glomerular deposits stain for IgG (most commonly IgG₁) and C3 and exhibit LC restriction in most cases. Electron microscopy is mandatory for diagnostic confirmation, showing glomerular deposition of microtubules of 10 to 90 nm in external diameter, with a distinct hollow core at magnification of less than 50,000, and arranged at least focally in parallel arrays ([Fig. 28.1](#)).^{24–27}

As in all glomerulopathies with monotypic Ig deposits, hematologic investigations searching for an underlying monoclonal B-cell clone are required. Two-thirds of patients have detectable serum and/or urine monoclonal gammopathy matching kidney deposits. ITGP/GOMMID complicates CLL or small lymphocytic lymphoma in about half of cases, and MGRS in the remaining cases.^{24–27} Intracytoplasmic microtubules, identical to those found in glomeruli, are commonly observed in clonal lymphocytes from these CLL patients ([Fig. 28.1](#)).^{24,27}

Data regarding treatment and outcomes in ITGP/GOMMID are limited. Kidney response to clone-targeted chemotherapy, using alkylating agents, rituximab-based, or bortezomib-based regimens, appears to be frequent and often sustained. When it occurs, renal relapse often precedes hematologic recurrence.^{26,27}

GLOMERULOPATHIES WITH NONORGANIZED MONOCLONAL IMMUNOGLOBULIN DEPOSITS

Glomerulopathies with nonorganized monoclonal immunoglobulins include two distinct entities: Randall-type MIDD and PGNMID.

Randall-Type Monoclonal Immunoglobulin Deposition Disease

Monoclonal Ig deposition disease encompasses three subtypes, depending on the composition of deposits. LCDD, in which the deposits contain monoclonal LC only (κ in 80% of cases) is by far the most common subtype. Heavy chain deposition disease (HCDD), with deposits derived from monoclonal heavy chains only (most frequently γ and always lacking the first constant [CH1] domain) is rare. Light and heavy chain deposition disease (LHCDD), in which the deposits are composed of monoclonal light and heavy chains (most commonly γ and κ) is even rarer.^{9,28–31}

Epidemiology

Monoclonal Ig deposition disease occurs with a slight male predominance and a wide range of age distribution, but predominantly in the sixth decade. Its diagnosis is made in around 0.3% to 0.5% of native kidney biopsies, representing a 10-fold lower prevalence compared with AL amyloidosis.^{28,29}

Clinical Manifestations

Monoclonal Ig deposition disease is due to MGRS in around two-thirds of cases but noticeably complicates symptomatic myeloma in more than 30% of cases. Abnormal LFC level and κ/λ ratio is a nearly constant finding in all types of MIDD.⁹ Monoclonal Ig deposition disease rarely occurs during WM, CLL, or non-Hodgkin lymphoma.^{9,29}

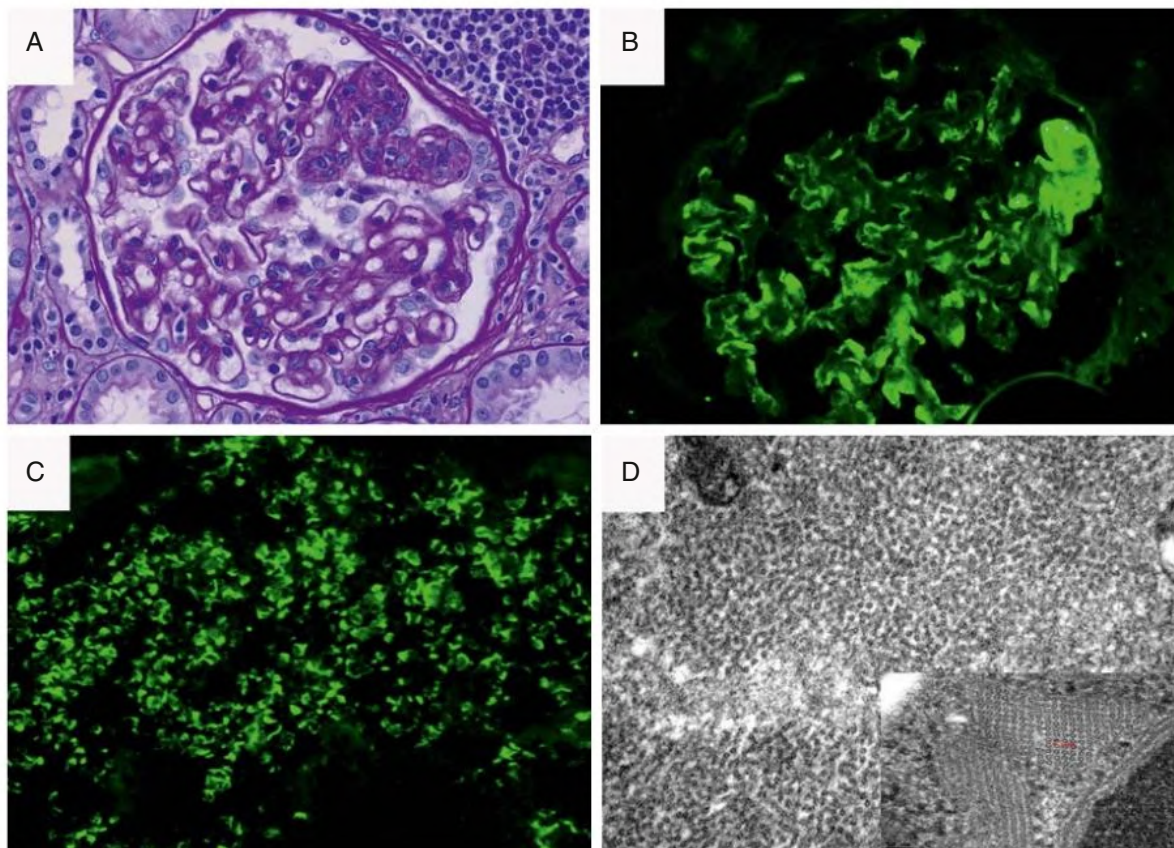


Fig. 28.1 Immunotactoid Glomerulopathy/Glomerulonephritis With Organized Microtubular Monoclonal Immunoglobulin (Ig) Deposits in a Patient With Underlying Chronic Lymphocytic Leukemia. (A) Endocapillary proliferative glomerulonephritis with segmental double contour appearance of capillary walls. Interstitial infiltration with malignant B cells (PAS $\times 400$). (B) Immunofluorescence microscopy. Glomerular granular deposits stained with anti-IgG2 and anti- κ (not shown) antibodies ($\times 400$). (C) Interstitial malignant B cells showing same positivity with anti-IgG2 antibody ($\times 400$). (D) Ultrastructurally, glomerular deposits are organized into microtubules of 10 nm in external diameter, with a distinct hollow core (electron microscopy $\times 100,000$). *Inset*: similar microtubules within the endoplasmic reticulum of clonal B-lymphocytes (electron microscopy $\times 100,000$).

Kidney involvement is an almost constant feature of MIDD. Heavy proteinuria is present in about 90% of cases, with hematuria and hypertension in around half of the patients. Nephrotic syndrome is more frequent in HCDD. High prevalence, early appearance, and rapid progression of CKD are other characteristics of kidney disease in MIDD.^{9,28-31} Around 10% of LCDD cases have prominent kidney arteriolar involvement and present with slowly progressive CKD and low-grade proteinuria (<0.5 g/day).⁹

Extrarenal monoclonal Ig deposition is frequent, particularly in LCDD. Liver, cardiac, and peripheral nerve manifestations have been reported in up to 30% of patients. Liver deposits may be confined to the sinusoids and basement membranes of biliary ducts, without associated parenchymal lesions. They are sometimes massive, with dilation and ruptures of sinusoids, resembling peliosis. Hepatomegaly with mild isolated cholestasis is the main presentation. Rarely, life-threatening hepatic insufficiency and portal hypertension may occur. Cardiac involvement results in hypertrophic cardiomyopathy, resembling AL amyloidosis cardiopathy, with potential development of diastolic dysfunction, arrhythmias, conduction blocks, and heart failure. Neuropathy due to deposits along peripheral nerve fibers or in the choroid plexus has been described. Deposits may affect a variety of other organs and tissues, including spleen, pancreas, salivary glands, thyroid, gastrointestinal tract, lungs, and skin.⁹

Kidney Pathology

By LM, the most characteristic lesion is nodular mesangial sclerosis, resembling diabetic nodular glomerulosclerosis (see [Chapters 29 and 31](#)), found in two-thirds of patients with LCDD and in almost all patients with HCDD. Mesangial nodules are composed of periodic acid–Schiff–positive material, with bitonal appearance, often accompanied by mild mesangial hypercellularity. Occasionally prominent endocapillary cellularity and mesangial interposition are present, mimicking MPGN. Crescents are uncommon, except in α -HCDD.^{9,30} Milder forms of LCDD may feature increased mesangial matrix or cellularity, with modest thickening of glomerular basement membranes (GBMs), often with abnormal brightness and rigidity. In early stages, glomerular deposits may be detected only by IF or EM.^{9,28-30}

The most characteristic and almost constant feature of MIDD is the presence of tubular lesions, predominantly in distal tubules, and characterized by the deposition of a refractile eosinophilic, ribbon-like material along tubular basement membranes (TBM). Deposits are commonly found in arterioles and peritubular capillaries. Variable degrees of tubular atrophy and interstitial fibrosis are present.^{9,28-30}

Immunofluorescence study is the cornerstone of diagnosis, which relies on the demonstration of diffuse linear TBM staining of monocytic LC only (LCDD), heavy chain only (HCDD), or both (LHCDD).

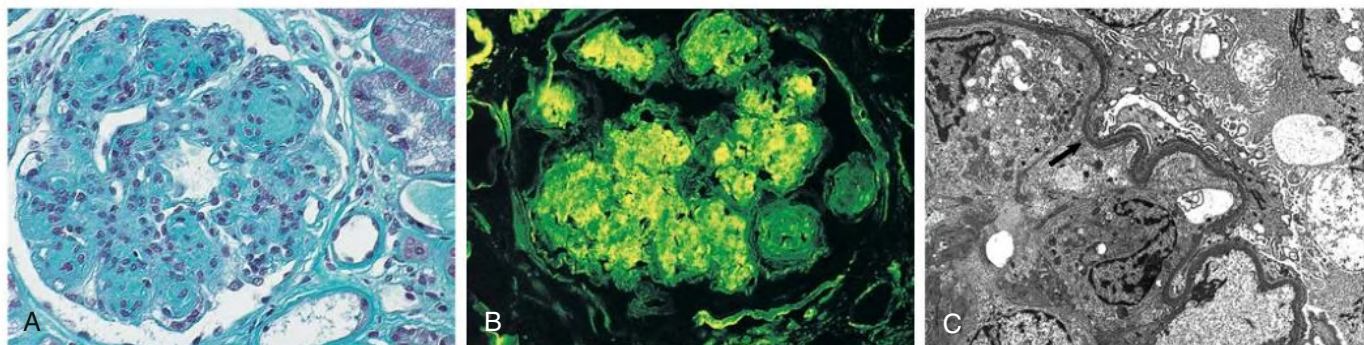


Fig. 28.2 Light Chain Deposition Disease. (A) Nodular glomerulosclerosis with mesangial matrix accumulation (Masson trichrome stain, $\times 312$). (B) Staining of mesangial nodules and tubular basement membranes with anti- κ antibody (immunofluorescence, $\times 312$). (C) Electron micrograph showing a layer of dense granular deposits (arrow) along the inner aspect of the glomerular basement membrane ($\times 2500$). (Courtesy Dr. Béatrice Mougenot, Paris.)

Similar deposits are common along GBMs, within the mesangium, and around vascular myocytes (Fig. 28.2).^{9,28-30} In γ -HCDD, diagnostic confirmation requires not only the absence of staining for κ and λ LCs, but also confirmation of monotypic γ -heavy chain deposits using antibodies specific for IgG subclasses. In addition, the use of antibodies specific for the constant domains invariably demonstrates deletion of the first (CH1) domain in HCDD. In most cases of $\gamma 1$ and $\gamma 3$ HCDD, codeposition of complement components C4 and C1q is present, associated with activation of the complement classical pathway in the serum.^{9,30}

By EM, the most characteristic feature is the presence of linear, finely granular punctate (“powdery”) electron-dense deposits along the inner aspect of GBMs and the outer aspect of TBMs.^{9,28-30}

A pattern of LCDD may be observed in patients with symptomatic myeloma and LC cast nephropathy. In this situation, glomerular lesions are typically absent, and linear deposits along glomerular and tubular basement membranes are identified by IF, defining the so-called LCDD by IF only.²⁸ LCDD with concurrent glomerular AL amyloidosis has been also described.⁹

Management and Outcomes

The outcome of patients with MIDD has improved with earlier diagnosis and development of efficient chemotherapy. In LCDD, before the era of novel antiplasma cell agents, kidney and overall median survivals were 2 and 4 years, respectively.²⁸ Currently 80% of patients are dialysis-free at 2 years from diagnosis. Median overall survival is estimated at more than 10 years in patients achieving hematologic response, who account for more than 70% of cases. Predictors of kidney response and survival include baseline GFR greater than 30 mL/min/1.73 m² and achievement of very good partial response (difference between involved and noninvolved sFLC <40 mg/L). In addition to hematologic and renal responses, factors influencing patient survival are age, presence of symptomatic myeloma, and extrarenal LC deposition.^{9,31}

Proliferative GN With Monoclonal Immunoglobulin Deposits

Epidemiology

PGNMID is a rare disorder with biopsy incidence around 1%. The mean age at presentation is 55 years, but PGNMID more commonly affects young patients than the other MGRS-related glomerulopathies.³²⁻³⁵

Clinical Manifestations

Proliferative GN with monoclonal Ig deposits is a renal-limited disorder, without any known description of extrarenal monoclonal Ig deposits. Kidney manifestations are nonspecific, with constant proteinuria and full-blown nephrotic syndrome in nearly half of patients. Hematuria, sometimes macroscopic, is found in 80% of cases. Two-thirds of patients have reduced GFR, but less than 10% require dialysis at the time of diagnosis. Rapidly progressive GN is uncommon. Low C3 and/or C4 serum levels are observed in around 20% of patients, and tests for cryoglobulins are negative.³²⁻³⁵

Diagnosis

Histologically, PGNMID mimics immune complex GN, with the association of proliferative glomerular lesions and deposits of immunoglobulins and complement.³²⁻³⁵ Predominant LM patterns are MPGN, endocapillary proliferative GN, or mesangial proliferative GN. In contrast to immune complex-type GN, glomerular deposits on IF appear monotypic, with staining for single LC isotype and a single γ -heavy chain subclass, admixed with C3 and C1q. There is a striking overrepresentation of the IgG3 κ isotype, particularly in young patients. The deposits are restricted to glomeruli (without extraglomerular or extrarenal involvement). PGNMID rarely features deposits made of other isotypes than IgG, including IgM, IgA, or LCs only.^{5,32,35} Electron microscopy shows granular, nonorganized deposits, typically in a subendothelial and mesangial distribution, with or without subepithelial deposits (Fig. 28.3).³²⁻³⁵

In contrast with most types of MGRS-related nephropathies, an underlying clone is infrequent, particularly in patients with IgG₃ deposits.³²⁻³⁵ A monoclonal Ig is found in around 30% using immunofixation and sFLC measurement. The diagnostic rate of bone marrow studies is also low, even using sensitive techniques such as flow cytometry.^{33,35} Accordingly, whereas some cases of PGNMID are actually MGRS-related, others, particularly those featuring IgG₃ deposits, might involve oligoclonal production of nephrotoxic Ig, secondary to skewed B-cell repertoire induced by viral or other putative antigenic stimulation.

Management and Outcomes

The specific management of PGNMID remains unclear, currently based on data from small cohorts. The frequent absence of a detectable clonal disorder affects both the choice of treatment and evaluation of its efficacy, which in many cases relies on kidney response only. Renin-angiotensin system blockade alone can be a reasonable strategy

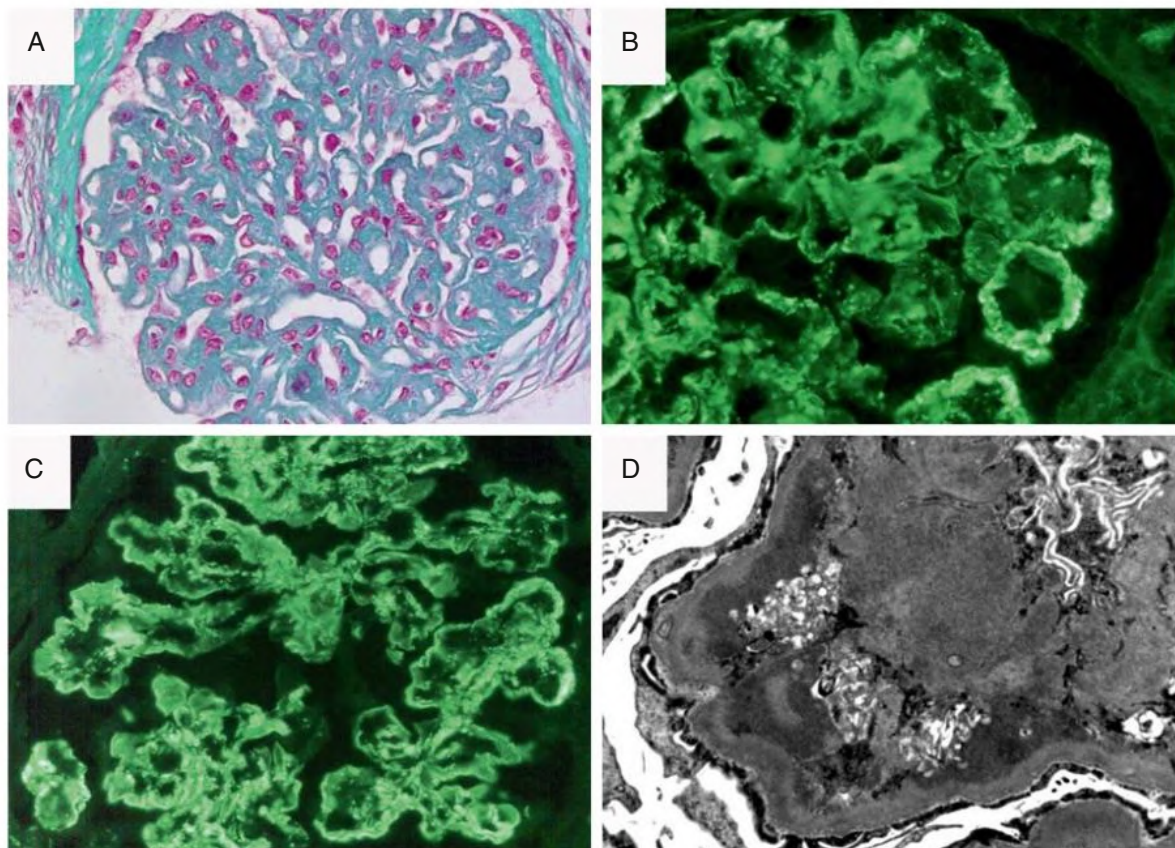


Fig. 28.3 Proliferative Glomerulonephritis (GN) with Monoclonal Immunoglobulin (Ig) Deposits. (A) Mesangial proliferative GN (trichrome staining $\times 400$). (B–C) Immunofluorescence microscopy showing glomerular granular deposits stained with anti-IgG3 (B) and anti- κ LC (light chain) (C) antibodies ($\times 400$). (D) By electron microscopy, discontinuous electron dense deposits appear nonorganized and have predominant subendothelial distribution ($\times 10,000$).

in patients without nephrotic syndrome and stable mild CKD, particularly if they are frail.^{10,32} Because PGNMID is a renal-limited disease, chemotherapy is not indicated in patients who have reached ESKD and are not candidates for kidney transplantation. In other situations, chemotherapy may be recommended, adapted to the underlying clone when identified, or empirically chosen and most often rituximab or bortezomib based.^{32,35}

RENAL AMYLOIDOSIS

The term *amyloidosis* refers to a heterogeneous group of diseases due to extracellular deposition of a protein material in an abnormal fibrillar conformation. Amyloidosis is classified according to the nature of the precursor protein that composes the fibrils. To date, more than 30 different types have been described in humans. Amyloidosis is most often an acquired disorder resulting from overproduction of a normal precursor protein. It may also be an inherited condition deriving from an abnormal genetically modified precursor. Most types of amyloidosis are systemic disorders, often with predominant kidney involvement^{8,36} (Table 28.2).

Common Properties to All Types of Amyloidosis

Whatever the protein precursor, amyloid deposits are composed of randomly arranged, rigid, nonbranching aggregated fibrils of indefinite length. Each amyloid fibril is composed of 4 to 6 twisted protofilaments, each displaying the typical “cross- β ” structure, where

antiparallel β -sheets are perpendicular to the filament axis. This configuration accounts for the affinity for Congo red staining that colors all amyloid deposits in a sensitive and specific manner, with a characteristic yellow-green birefringence under polarized light. All types of amyloid fibrils are associated with common nonfibrillar components. Most important is the amyloid P component, a glycoprotein that represents up to 15% of deposits and accounts for their incapacity to be solubilized in an organic milieu. It derives from serum amyloid P protein (SAP), a member of the pentraxin family (as C-reactive protein), synthesized by the liver. Serum amyloid P protein is resistant to proteolytic digestion, and its coating to amyloid fibrils results in their reduced tissue catabolism.^{8,11} Another constant component of amyloidosis are glycosaminoglycans (GAGs), which are linear polysaccharides made of repeating units of hyaluronic acid. Linked to a protein core, they form proteoglycans, mostly of the heparan sulfate type. Serum amyloid P protein easily binds to GAGs, which appear to induce and stabilize the β -pleated amyloid structure.^{8,36}

Amyloid formation results from abnormal folding of the precursor protein, which modifies its structure leading to the formation of intermolecular β -sheets. Amyloidogenesis involves a nucleation-dependent polymerization process. Formation of an ordered nucleus is the initial and thermodynamically limiting step, followed by addition of monomers and elongation of the fibrils. This process depends on (1) the natural propensity of the precursor to adopt an abnormal configuration (e.g., chronic accumulation of native transthyretin with aging, leading to wild-type transthyretin amyloidosis); (2) overproduction or

TABLE 28.2 Main Characteristics of Systemic Amyloidoses

Type	Acquired or Hereditary	Underlying Disorder	Precursor Protein	ORGAN INVOLVEMENT					Treatment
				Kidneys	Heart	Liver	PN (AN)	Other	
AL, AH	Acquired	Plasma cell clone (usually MGR(CIS))	Monoclonal Ig LC Monoclonal Ig heavy chain	+++	+++	++	++(+)	Soft tissue, GI	Chemotherapy ± anti-CD38 Mab HDM/ASCT
AA	Acquired	Chronic inflammatory disorders (infections, chronic arthritis, FMF, other periodic fever syndromes)	SAA	+++	± (late)	+	(late)	GI, adrenal glands, thyroid	Suppression of inflammation Anti-TNF and anti-IL1 agents Colchicine
ATTR	Acquired	Aging	Wild-type TTR	–	+++	–	–	Carpal tunnel syndrome	Supportive treatment of heart failure Tafamidis
	Hereditary	Mutations in <i>TTR</i> gene	TTR variant	+	++	–	+++(+++)	Eye	Supportive treatment of heart failure Tafamidis, RNA interference Liver transplantation (in selected patients)
Aβ2M	Acquired	Long-term dialysis	Wild-type Aβ2M	–	–	–	–	Arthropathy Carpal tunnel	Efficient hemodialysis, renal transplant
	Hereditary	Mutation in Aβ2M gene	Aβ2M variant	–	+	+	++(+)	GI, salivary glands, spleen, adrenal glands	Supportive
ALECT2	Acquired	Unknown	Wild-type Lect2	+++	–	++	–	–	Supportive
Afibr	Hereditary	Mutations in fibrinogen α chain gene	Fibrinogen Aα chain variant	+++ (glomerular)	–	±	–	Spleen	Supportive, organ transplant
ALys	Hereditary	Mutations in lysozyme gene	Lysozyme variant	+	–	++	–	GI, spleen, skin, salivary glands	Supportive
AGel	Hereditary	Mutations in gelsolin gene	Gelsolin variant	±	–	–	++ cranial (–)	Eye, skin	Supportive
AApoA1	Hereditary	Mutations in apolipoprotein A1 gene	ApoA1 variant	++	+	++	± (–)	Testis, larynx, skin, eye	Supportive, organ transplant
AApoA2	Hereditary	Mutations in apolipoprotein A2 gene	ApoA2 variant	+++	–	–	–	–	Supportive
AApoC2	Hereditary	Mutations in apolipoprotein C2 gene	ApoC2 variant	+++	–	–	–	–	Supportive
AApoC3	Hereditary	Mutations in apolipoprotein C3 gene	ApoC3 variant	+++	–	+	–	GI, salivary glands, bronchial	Supportive
ALm	Hereditary	Mutation in κ LC constant domain gene	Constant κ LC domain	+++	+	+	–	GI, thyroid, pancreas	Supportive

AL, Heavy chain amyloidosis; AH, light chain amyloidosis; FMF, familial Mediterranean fever; GI, gastrointestinal; HDM/ASCT, high-dose melphalan/autologous stem cell transplantation; Ig, immunoglobulin; IL-1, interleukin-1; LC, light chain; MGR(CIS), monoclonal gammopathy of renal (clinical) significance; SAA, serum amyloid A; TNF, tumor necrosis factor. Modified from Wechalekar AD, Gillmore JD, Hawkins PN. Systemic amyloidosis. *Lancet*. 2016;387:2641–2654.

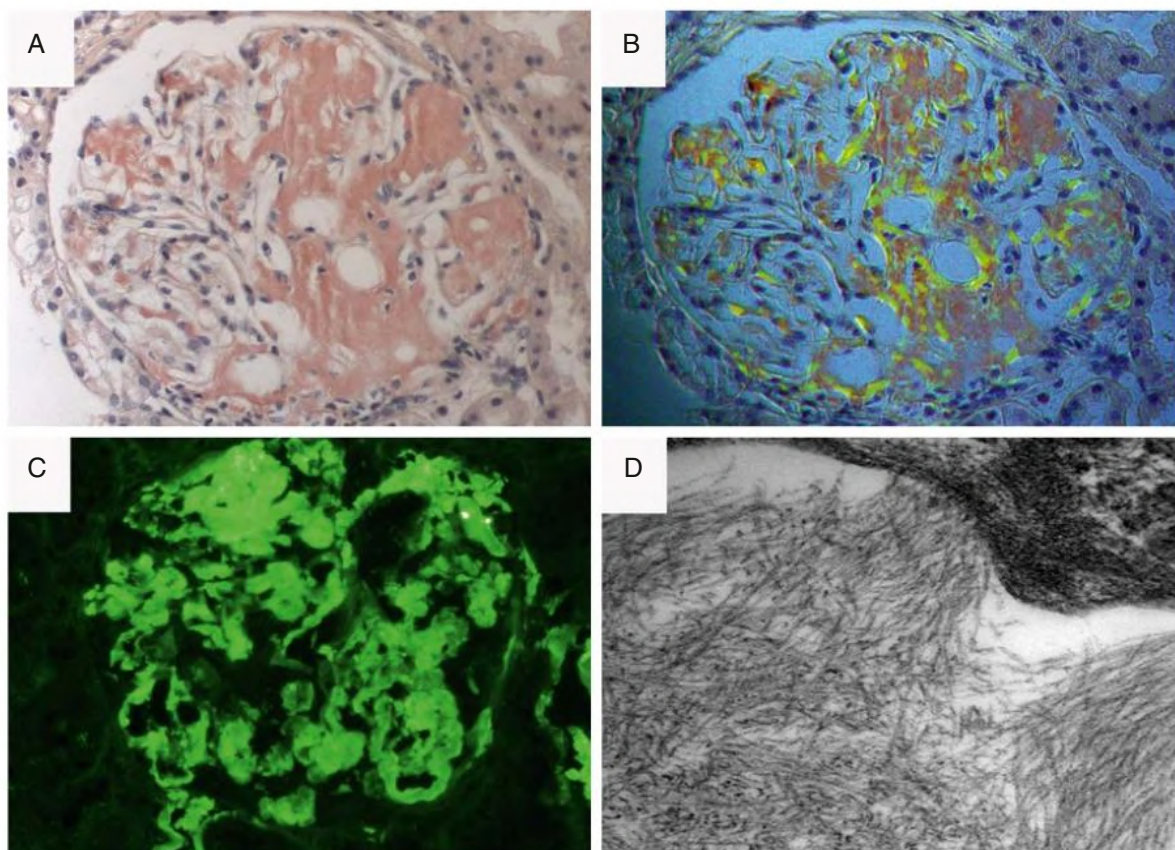


Fig. 28.4 Light Chain Amyloidosis. (A) Glomerular mesangial and capillary wall Congo red–positive deposits (Congo red staining, $\times 400$). (B) Glomerular amyloid deposits showing dichroism and birefringence under polarized light (Congo red staining, $\times 400$). (C) Immunofluorescence microscopy: positive staining with anti- λ antibody ($\times 400$). (D) Electron microscopy: randomly arranged unbranched amyloid fibrils of 7 to 10 nm in external diameter ($\times 100,000$).

accumulation of the precursor (e.g., serum amyloid A protein [SAA] or β -2 microglobulin, resulting in acquired amyloid A [AA] amyloidosis or β -2 microglobulin amyloidosis, respectively); (3) genetic mutation generating or increasing amyloidogenicity of a precursor protein in hereditary amyloidosis; and (4) modification of the precursor through proteolytic cleavage or other mechanisms. In AL amyloidosis the LC variable domain (VL) is the main component of the fibrils, its polymerization being favored by proteolysis of the constant domain. In AA amyloidosis, macrophages seem to enhance fibril formation through C-terminal proteolysis of the precursor SAA.^{8,36}

Diagnosis of Amyloidosis

General Considerations

Amyloidosis should be considered in any patient with suggestive presentations, including heart failure with preserved ejection fraction, albuminuria, or peripheral neuropathy, particularly if a monoclonal gammopathy is present. However, this diagnosis requires histologic confirmation. Whenever possible, noninvasive biopsies of abdominal fat and minor salivary glands should be performed initially. If necessary (i.e., when tissue biopsies fail to demonstrate amyloid deposition or are insufficient for amyloid typing), biopsy of a clinically affected organ (usually the kidney) should be considered. The diagnosis of amyloid is based on light microscopic examination, showing amorphous extracellular Congo red–positive deposits (Fig. 28.4). Congo red staining may be falsely negative if tissue sections are less than 5 μ m in thickness. When a kidney biopsy is performed in patients with kidney

involvement, it allows identification of Ig LC amyloid deposits in more than 80% of cases.^{5,8,36}

Rapid identification of the nature of the amyloid precursor is crucial, as it dictates therapeutic management. It relies primarily on immunohistochemistry on paraffin-embedded tissue sections, or preferably on IF microscopy on frozen sections, using a panel of reactive antibodies that should include at least conjugates specific for κ and λ LC, Ig heavy chains, SAA, and transthyretin. Immunofluorescence is associated with higher rate of successful typing than immunohistochemistry (Fig. 28.4C). However, even IF has limitations due to local proteolysis, inaccessibility of the epitopes, or lack of commercial antibodies. Considerable progress has been achieved with the development of novel techniques, such as immuno-EM and laser capture microdissection combined with tandem mass spectrometry. Due to their high sensitivity and specificity, they allow correct typing in nearly 100% of cases. Although available only in a few specialized centers, they should be systematically considered when routine antibody-based techniques are inconclusive.^{37,38}

Kidney Pathology

On LM, extracellular eosinophilic deposits appear faintly red after Congo red staining (Fig. 28.4A) and show characteristic apple-green birefringence under polarized light (Fig. 28.4B). The earliest lesions are located in the mesangium (Fig. 28.4A), on the external aspect of GBMs, and in blood vessels. Mesangial deposits may be sparse or more diffuse, sometimes featuring a lobular distribution. When glomeruli

become massively sclerotic (or at the opposite, in the very early stages), the deposits may be difficult to demonstrate by Congo red staining. In this situation, EM may be useful to confirm the presence of amyloid deposits that typically display the ultrastructural appearance of randomly arranged fibrils 7 to 10 nm in external diameter (Fig. 28.4D).⁵

Main Types of Systemic Amyloidosis With Kidney Involvement

Acquired systemic AL, AA, and leukocyte cell-derived chemotaxin 2-associated amyloidosis (ALECT2) are the most frequent causes of amyloid kidney disease. Rarely, amyloidosis-associated kidney disease is inherited, induced by a mutation of the precursor protein (Table 28.1).

Clinical Presentation

Whatever its type, amyloidosis-associated kidney disease most commonly manifests with chronic glomerular symptoms, including heavy proteinuria (usually around 5 g/day at presentation) predominantly composed of albumin. Proteinuria is sometimes massive (>20 g/day), and about half of patients have full-blown nephrotic syndrome, often accompanied by diffuse edema. Fluid retention is of variable severity, sometimes resulting in anasarca. Nephrotic syndrome due to amyloidosis increases the risk of thromboembolic complications including renal vein thrombosis, particularly in AA amyloidosis. The prevalence of hematuria and hypertension varies according to the type of amyloidosis, being uncommon in AL whereas observed in around 30% in patients with AA. Kidney failure appears more frequent and usually more severe in AA compared with AL, reported in 50% to 75% and less than 50% in newly diagnosed patients, respectively.³⁹ Classically, enlarged kidneys on imaging studies are suggestive of amyloidosis, but kidneys of normal or even reduced size should not exclude the diagnosis, particularly in patients with kidney failure.^{15,35,36}

Rarely, amyloid deposits predominantly or exclusively involve the tubulointerstitial compartment, manifesting with low-grade tubular proteinuria, slowly progressive chronic kidney failure, and impaired urine concentration with polyuria. This misleading presentation has been mostly reported in AA and in AL amyloidosis, and more specifically in hereditary apoA1 amyloidosis related to the Leu75Pro variant.⁴⁰ Amyloid deposits may predominantly affect intrarenal vessels, inducing severe hypertension, as in apoC3 hereditary amyloidosis (Table 28.2).

Immunoglobulinic Amyloidosis (Light Chain and Heavy Chain)

In the vast majority of cases, the main component of amyloid fibrils derives from the VL of a monoclonal LC. Heavy chain (AH) amyloidosis, made of a peptide derived from an Ig heavy chain, is rare. The incidence of AL amyloidosis is 9 to 12 cases per million inhabitants per year. AL is always due to a LC-secreting clonal population of cells belonging to the B-cell lineage. The abnormal population rarely causes an overt lymphoid disorder, such as symptomatic MM or WM. In over 80% of cases, it corresponds to a MG(R)CS, as defined earlier.^{8,36} The plasma cell clone that usually characterizes these MGCS-related ALs is featured by similar cytogenetic abnormalities as in symptomatic MM, except for a striking overrepresentation of the translocation t(11–14) (more than 50% compared with 15%, respectively). This translocation is associated with relative resistance to apoptosis and may predict high efficacy of BCL2 inhibitors and less sensitivity to bortezomib and immunomodulatory drugs.⁸

Physicochemical and structural characteristics determining the amyloidogenic potential of monoclonal LC remain poorly understood. The λ LC isotype is up to fourfold more frequent than the κ isotype, whereas in other monoclonal gammopathies, κ accounts for two-thirds of cases. The tropism of organ involvement is influenced

TABLE 28.3 Main Clinical and Laboratory Features at Presentation in 816 Patients With Proven Light Chain (AL) Amyloidosis

Median age (years)	63
Male sex	59.9%
Organ involvement	
Median number of involved organs	2
Kidney	68.1%
Heart	64.8%
Peripheral nerves	18.7%
Liver	16.0%
eGFR value (mL/min/1.73 m ²)	
30–59	25.1%
15–29	8.5%
<15 or dialysis	6.9%
Median proteinuria (g/day)	2.9
NYHA class III or IV	26.8%
Cardiac staging	
I	30.9%
II	43.7%
III	25.4%
Involved LC type	
κ	24.6%
λ	75.4%
Positive serum/urine immunofixation	94.2%
Monoclonal gammopathy	
IgG	38.8%
IgA	10.0%
IgM	3.0%
LC only	47.6%
Serum FLCs	
Abnormal κ/λ ratio	84.8%
Median involved FLC concentration (mg/L)	178
Median dFLC (mg/L)	157
Median bone marrow plasma cell infiltration	10%

eGFR, Estimated glomerular filtration rate; NYHA, New York Heart Association; FLC, free light chain; dFLC, difference between involved and noninvolved FLC; Ig, immunoglobulin; LC, light chain.

Modified from Palladini G, Dispenzieri A, Gertz MA, et al. New criteria for response to treatment in immunoglobulin light chain amyloidosis based on free light chain measurement and cardiac biomarkers: impact on survival outcomes. *J Clin Oncol*. 2012;30:4541–4549.

both by the germline gene used for the light-chain VL and by somatic mutations occurring in the secreting clone. Thus, the usage of genes belonging to the rare V λ 6 family is constantly associated with kidney involvement, whereas κ LCs of the V κ 1 family are more likely to induce dominant hepatic involvement. In addition, organ-specific environmental factors are also involved. Independently of deposition as fibrils, circulating free amyloid LCs may also participate in organ damage, as documented in the heart and suspected in the kidney.^{6,8}

Clinical Manifestations

The median age at diagnosis is 64 years, with a slight male predominance (Table 28.3). About 10% of the patients are younger than 50 years. The clinical presentation of AL amyloidosis depends on the pattern and severity of organ involvement. Amyloid deposits may affect virtually any organ or tissue. Except for the combination of macroglossia (Fig. 28.5) and periorbital purpura (Fig. 28.6), which is rare but pathognomonic of AL amyloidosis, clinical features are not specific, with weakness being the most frequent initial symptom. Thus the



Fig. 28.5 Macroglossia in patient with light chain amyloidosis.



Fig. 28.6 Periorbital purpura in patient with light chain amyloidosis.

clinical diagnosis may be challenging, and when facing a patient with glomerular symptoms, it is crucial to carefully search for extrarenal manifestations.

Kidney disease is present in up to 70% of patients. Cardiac involvement is as frequent, and it is the major cause of mortality through arrhythmia, conduction blocks, and congestive heart failure. It is typically featured by restrictive hypertrophic cardiomyopathy, which should be systematically detected with measurement of sensitive cardiac biomarkers (troponin and brain natriuretic peptide [BNP] or NT-proBNP) and Doppler echocardiography showing increased septum ventricular wall thickness and decreased left ventricular longitudinal strain. Magnetic resonance imaging is useful in difficult cases. AL amyloidosis may infiltrate articular structures, such as carpal tunnel. Peripheral nerve involvement may result in a sensory polyneuropathy, often associated with dysautonomia including orthostatic hypotension, impotence, and bladder dysfunction. Involvement of the liver, spleen, and gastrointestinal tract is common and usually pauci-symptomatic. The most common skin manifestation is purpura, secondary to amyloid infiltration of capillaries. This implies a hemorrhagic risk, sometimes increased by an acquired deficiency of Factor X, thought to occur

through adsorption on amyloid fibrils. Accordingly, coagulation and hemostasis screening tests are particularly important before biopsy of the kidney or any other deep organ.^{8,36}

Management and Outcomes

The diagnostic workup of AL amyloidosis is similar to that of other MG(R)CS (see earlier). As already mentioned, the presence of a monoclonal gammopathy is not sufficient for the diagnosis, which should be secured by histologic identification of the Ig nature of amyloid deposits. AL amyloidosis is a systemic disease, and once the diagnosis has been confirmed, biopsies of other potentially involved sites are unnecessary and should be avoided because of the bleeding risk.

Elevation of NT-proBNP and cardiac troponin strongly correlate with prognosis and are therefore used for risk assessment according to the Mayo staging system. This prognostic score identifies 3 stages: stage 1, normal serum levels of troponin T and NT-proBNP; stage 2, increase in a single marker level; and stage 3, increase in both markers with thresholds at 332 ng/L for NT-proBNP and 0.035 $\mu\text{g/L}$ for troponin, respectively.^{8,16,36} Of note, NT-proBNP serum levels are influenced by eGFR. This limitation can be partly overcome by using BNP in patients with severe kidney failure.

Patients presenting with very high concentrations of NT-proBNP (>8500 ng/L) with systolic hypotension less than 100 mm Hg (Mayo Clinic stage IIb) have a median survival of only 4 months. Cardiac transplantation may be considered in selected patients. Kidney involvement has a lesser impact on survival.⁸ Proteinuria greater than 5 g/24 h with eGFR of 50 mL/min/1.73 m² or less at diagnosis have been associated with a high risk for progression to ESKD (up to 85% at 3 years).¹⁶

Regarding clone-targeted chemotherapy, pioneering studies established a modest efficacy of the oral melphalan and prednisone (MP) regimen. An important progress was achieved by using high-dose dexamethasone instead of prednisone (MDex combination), which significantly increased both frequency and rapidity of hematologic response. From the early 1990s, HDM/ASCT has been performed in selected patients with AL. Even in reference centers and after careful patient selection, HDM/ASCT is associated with significant morbidity and an early mortality rate of around 5%.^{8,36} A randomized controlled trial performed in a multicenter setting did not provide evidence for a survival benefit of HDM/ASCT compared with MDex.⁴¹ However, some centers still offer HDM/ASCT to highly selected young Mayo stage 1 individuals, who represent a minority of AL patients. Other investigators consider that the results of HDM/ASCT are biased by the selection process.

A decisive step has been the introduction of novel antimyeloma agents, particularly the proteasome inhibitor bortezomib. To date, bortezomib-melphalan-dexamethasone (BMDex) or cyclophosphamide-bortezomib-dexamethasone (CyBorD) regimens are the backbone of initial treatment.¹³ The use of antiplasma cell monoclonal antibodies, such as anti-CD38 Mabs, is likely to further increase hematologic response rate, which remains the key prognostic factor.¹⁵ Results of a recent phase III trial indicate that addition of subcutaneous daratumumab to CyBorD results in hematologic response rate of more than 90%, compared with 77% with CyBorD alone. In addition, deep responses were significantly more frequent in the daratumumab arm, and this translated into increased organ improvement.⁴²

Hematologic response should be regularly evaluated to assess treatment efficacy and to rapidly detect disease relapse. It is based on repeat sFLC measurements, using the difference between serum level of involved and noninvolved light chains (dFLC). Current response criteria are defined as complete response (negative serum and urine

immunofixation with normalized FLC ratio), very good partial response (reduction of dFLC <40 mg/L), and partial response (50% reduction of dFLC). Regarding organ response, the modification of serum NT-ProBNP and troponin levels is an important prognostic factor in patients with cardiac amyloidosis. Because improvement in kidney parameters is usually delayed by several months following hematologic remission, treatment modifications should be based on FLC response.^{8,16–19} Although several agents developed to increase clearance of amyloid deposits have failed to demonstrate a clinical benefit,¹¹ this approach is still under evaluation, particularly using novel antifibril antibodies.^{8,12}

AA Amyloidosis

General Considerations

AA amyloid fibrils derive from SAA, which is produced by the liver during inflammatory reaction, where its serum concentration may increase by a factor of 1000. Among the different SAA isotypes, whose function is unknown, only SAA1 and SAA2 are amyloidogenic.

AA amyloidosis is consecutive to chronic inflammation, either of infectious origin or due to chronic inflammatory diseases, most commonly acquired and represented by rheumatoid disorders. AA amyloidosis sometimes complicates a genetic disorder due to mutation in a factor involved in the control of inflammatory reaction. These disorders, referred to as hereditary periodic fever syndromes, are increasingly recognized as the underlying condition in AA amyloidosis.

Patients at risk for of AA amyloidosis are those with long duration of chronic inflammatory disease (median 17 years), high serum SAA levels, homozygosity for the SAA1 genotype, familial Mediterranean fever (FMF) trait (heterozygosity for variant pyrin), or other family history of AA amyloidosis (e.g., periodic fever syndromes).^{39,43}

AA Amyloidosis Due to Acquired Inflammation

AA amyloidosis is rare in high-income countries, where it mostly complicates chronic inflammatory arthritis, particularly rheumatoid arthritis and systemic juvenile idiopathic arthritis, less commonly poorly controlled inflammatory bowel disease, or chronic infections such as in bronchiectasis and osteomyelitis. Rarely, AA amyloidosis complicates neoplasia, Castleman disease, and WM. AA is still frequent in lower-income countries and is commonly observed during granulomatous infections such as tuberculosis and pyogenic infections, particularly chronic osteomyelitis.

The usual clinical presentation of these forms of AA amyloidosis is proteinuria, often in the nephrotic range, with or without reduced glomerular filtration rate, in the context of a chronic infectious process or poorly controlled inflammatory arthritis.

Compared with AL amyloidosis, kidney disease in AA amyloidosis is more frequent and severe and is the main prognostic factor for survival. In contrast, symptomatic cardiac involvement is observed in less than 10% of cases. Gastrointestinal tract infiltration, hepatomegaly, and spleen enlargement are common, often asymptomatic, but hyposplenism may occur resulting in infectious complications. Marked cholestasis is rare, but associated with poor outcomes. Endocrine manifestations, including adrenal and thyroid dysfunction may occur. Peripheral neuropathy is rarely reported (Table 28.4).

Amyloid load and clinical outcomes relate to circulating concentrations of SAA. The risk for death among patients with an SAA concentration of 4 mg/L or less is almost 18 times lower than in those with an SAA concentration of 155 mg/L or greater. Even a very modest elevation in the SAA concentration of 4 to 9 mg/L is associated with a fourfold increased risk for death. These data emphasize the importance of vigorous treatment of the underlying inflammatory disease.

TABLE 28.4 Characteristics at Presentation of 374 Patients With Systemic Secondary (AA) Amyloidosis

Age, years (range)	50 (9–87)
Male sex	210 (56%)
Race or Ethnic Group	
White	307 (82%)
South Asian	27 (7%)
Other	40 (11%)
Duration of Inflammatory Disease at Diagnosis (Years)	
Median	17
Range	0–68
Kidney Dysfunction	
Proteinuria >500 mg/day or serum creatinine >133 μmol/L	363 (97%)
End-stage kidney disease	41 (11%)
Proteinuria, g/day median (range)	3.9 (0–26.0)
Liver Involvement	
Hepatomegaly, n (%)	35 (9%)
Deposits on SAP scintigraphy	85 (23%)
Splenic Involvement	
Deposits on SAP scintigraphy	370 (99%)
Cardiac Involvement	
Cardiac failure (n)	1
Cardiac infiltration (n)	2

From Lachmann HJ, Goodman HJ, Gilbertson JA, et al. Natural history and outcome in systemic AA amyloidosis. *N Engl J Med*. 2007;356:2361–2371.

Antibiotic treatment or surgical resection of an infectious site efficiently prevents and may even cure infection-related AA amyloidosis by suppressing its cause. The increasing use of effective disease-modifying treatments for chronic inflammatory arthritis, such as anti-tumor necrosis factor and anti-interleukin-1 agents, accounts for the decline of AA in developed countries.³⁹

AA Amyloidosis in Autoinflammatory Syndromes

Familial Mediterranean fever, also referred to as periodic fever, is the most frequent auto-inflammatory syndrome complicated by AA amyloidosis. Familial Mediterranean fever is an autosomal recessive disorder occurring most often in Sephardic Jews and Armenians. It is caused by mutations of the Mediterranean fever (MEFV) gene encoding a protein called pyrin (also known as marennostin). Clinically, there are two independent phenotypes. In the first, brief episodic, febrile attacks of peritonitis, pleuritis, or synovitis occur in childhood or adolescence and precede the kidney manifestations. In the second phenotype, kidney symptoms appear first and may long be the only manifestation of the disease. The attacks are accompanied by dramatic elevations of acute phase reactants, including SAA. Amyloid deposits are responsible for severe kidney lesions with prominent glomerular involvement, leading to ESKD at a young age. When given early, colchicine can prevent the development of proteinuria. It is less effective in preventing eGFR decline in patients with nephrotic syndrome or kidney impairment. The minimum daily dose of colchicine for prevention of amyloidosis is 1 mg, but doses of 1.5 to 2 mg are required in patients with symptomatic amyloid nephropathy. In patients unresponsive or intolerant to colchicine, or noncompliant, interleukin-1 receptor antagonists should be considered.⁴³

Other autoinflammatory syndromes associated with specific genetic mutations have been identified. All may be complicated by

AA amyloidosis. The main are TNF- α receptor-associated periodic fever syndrome (TRAPS), Muckle-Wells syndrome, familial cold autoinflammatory syndrome, and hyperIgD syndrome. Most can be efficiently controlled by anti-interleukin-1 or anti-tumor necrosis factor- α agents.^{39,43}

Other Types of Amyloidosis With Kidney Involvement

Leukocyte Cell-Derived Chemotaxin 2-Associated Amyloidosis

Recently identified, ALECT2 is currently the third most common type of kidney amyloidosis in the United States. It preferentially affects certain ethnic groups, particularly Hispanic Americans. LECT2 is a chemotactic factor to neutrophils mainly produced by the liver. ALECT2 manifests around 60 years of age, without family history. Kidney disease is predominant, including proteinuria and CKD that progresses to ESKD in 40% of cases. Nephrotic syndrome is rare. Kidney deposits predominate in the cortical interstitium, with variable glomerular and vascular distribution. Deposits may involve the liver and adrenal glands. The pathophysiology remains unknown. There is no mutation in LECT2 but affected individuals show homozygous polymorphism for the G allele, suggesting the disease may be favored by environmental factors on a predisposing genetic background.⁴⁴

Hereditary Kidney Amyloidoses

The kidney is the major site of organ involvement in most types of hereditary amyloidosis, including fibrinogen A- α , apolipoprotein A-I, apolipoprotein A-II, apolipoprotein C2, apolipoprotein C3, gelsolin, and lysozyme amyloidosis. The exception is hereditary transthyretin amyloidosis (ATTR), the most frequent type of genetically transmitted amyloidosis, in which cardiac disease and peripheral neuropathy with

dysautonomia are prominent, whereas proteinuria and kidney failure develop in 30% to 50% of patients only. All hereditary amyloidoses are autosomal dominant conditions, but due to variable penetrance, a relevant family history may be absent. Thus, kidney hereditary amyloidosis should be suspected whenever IF studies of the kidney biopsy fail to type amyloid deposits. In this situation, the diagnosis is established by proteomic and/or immuno-EM completed by genetic studies to confirm an amyloidogenic mutation of the precursor protein. Careful clinical investigation of extrarenal manifestations is required to guide diagnosis and to evaluate organ involvement.³⁶

Fibrinogen A- β Amyloidosis

Fibrinogen A- α Amyloidosis (AFib) is the most frequent hereditary kidney amyloidosis, deriving from a single mutation in the fibrinogen A- α -chain gene. It clinically manifests with chronic glomerular disease, usually around age 58 years. A family history is often lacking, resulting in delayed diagnosis. Age at diagnosis and severity of presentation vary according to the type of mutation. Symptomatic extrarenal manifestations are uncommon, although liver and spleen involvement may occur. In the kidney, amyloid deposits have a characteristic distribution, because they massively infiltrate the glomerulus but almost always spare the interstitial and vascular compartments. Kidney disease progresses to ESKD after a median time of 4.6 years. Kidney transplantation is usually associated with acceptable graft survival, despite disease recurrence. In the most severe forms (juvenile onset, hepatic failure) combined liver and kidney transplantation should be considered because it prevents amyloid recurrence by removing the source of the amyloidogenic fibrinogen variant, taking into account the increased risk of perioperative complications.⁴⁵

SELF-ASSESSMENT QUESTIONS

- Which of the following applies to heart involvement in AL (light chain) amyloidosis?
 - A contraindication to cardiac transplantation
 - Less common than in AA amyloidosis
 - Primarily responsible for valve dysfunction
 - A contraindication to hemodialysis
 - A major prognostic factor for patient survival
- Regarding renal disease in monoclonal immunoglobulin deposition disease (MIDD), which of the following is *true*?
 - A contraindication to kidney transplantation
 - Characterized by crescent formation
 - Frequently responsible for nondiabetic glomerulosclerosis
 - Characterized by microtubular deposits
 - Caused by deposition of cryoglobulin
- Which of the following is the key investigation for the diagnosis of immunotactoid glomerulopathy?
 - Immunofixation of serum proteins
 - Immunofixation of urinary proteins
 - Bone marrow aspiration
 - Immunofluorescence examination of kidney biopsy sample
 - Electron microscopy
- Which of the following is the *most* common clinical sign or symptom caused by MIDD?
 - Renal failure
 - Hematuria
 - Hypertension
 - Jaundice
 - Heart failure

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